

What is Human Physiology?

the study of biological function of
humans

Levels of Structural Organization in the Human Body

- Chemicals
- Cells
- Tissues
- Organs
- Organ
Systems

The Chemical Level

- Human body is composed of *atoms* organized into *molecules*
- Atoms and molecules undergo *chemical reactions*
 - Physiology involves control of chemical reactions within the body

Cells

- Basic living units
- Smallest subdivision able to carry out all life processes
- Contain *organelles*
 - Structures that have specific functions
- Specialized for specific physiological roles

Tissues

- Groups of cells of similar structure
- Interaction among cells leads to functions single cells cannot effectively do alone

Types of Tissues

- Epithelial tissues
 - Form body surfaces
 - Barriers
 - Secretion
 - Absorption
 - Some movement (cilia)

Types of Tissues

- Connective Tissues
 - Very diverse
 - *Extracellular matrix*
 - Incorporate large amounts of extracellular material in tissue structure and function
 - Connection, Structure, and Protection

Types of Tissues

- Muscle Tissue
 - Contraction
 - Generates Tension, Movement and Heat

Types of Tissues

- Nerve tissue
 - High speed communication
 - Control/Integration

Organs

- structures consisting of at least two tissue types
- performs a specific function related to the whole body

Organ Systems

- groups of organs performing related functions
- Enable basic functions needed to maintain overall homeostasis

Major Organ Systems

- Nervous System
 - Communication and control
- Endocrine System
 - Communication and control
- Integumentary System
 - Barrier between interior of body and exterior, thermoregulation
- Respiratory System
 - Gas exchange, pH balance
- Urinary System
 - Waste removal, water and ion balance

Major Organ Systems

- Digestive System
 - Obtain raw materials from exterior
- Immune System
 - Protect body from foreign substances, other organisms and cancer cells
- Musculoskeletal System
 - Movement, structure, support, protection, heat generation
- Circulatory System
 - Transportation of most materials throughout body
- Reproductive System
 - Perpetuation of the species

The Internal Environment

- The interior of body, the environment of cells inside the body
- Mostly water (~67%)
 - Liquid
 - High heat capacity
 - does not change temperature easily
 - Polar solvent
 - dissolves some substances, not others

Attributes of the Internal Environment

- amt. nutrients
- amt. wastes
- amt. O₂ & CO₂
- amt. salts
- Temperature (37°C)
- pH (7.4)
- Fluid volume

Various physiological systems maintain different attributes of the internal environment within narrow normal ranges

Homeostasis

= state of constancy of conditions
within the body

How does the body maintain
homeostasis?

Through Regulatory Mechanisms

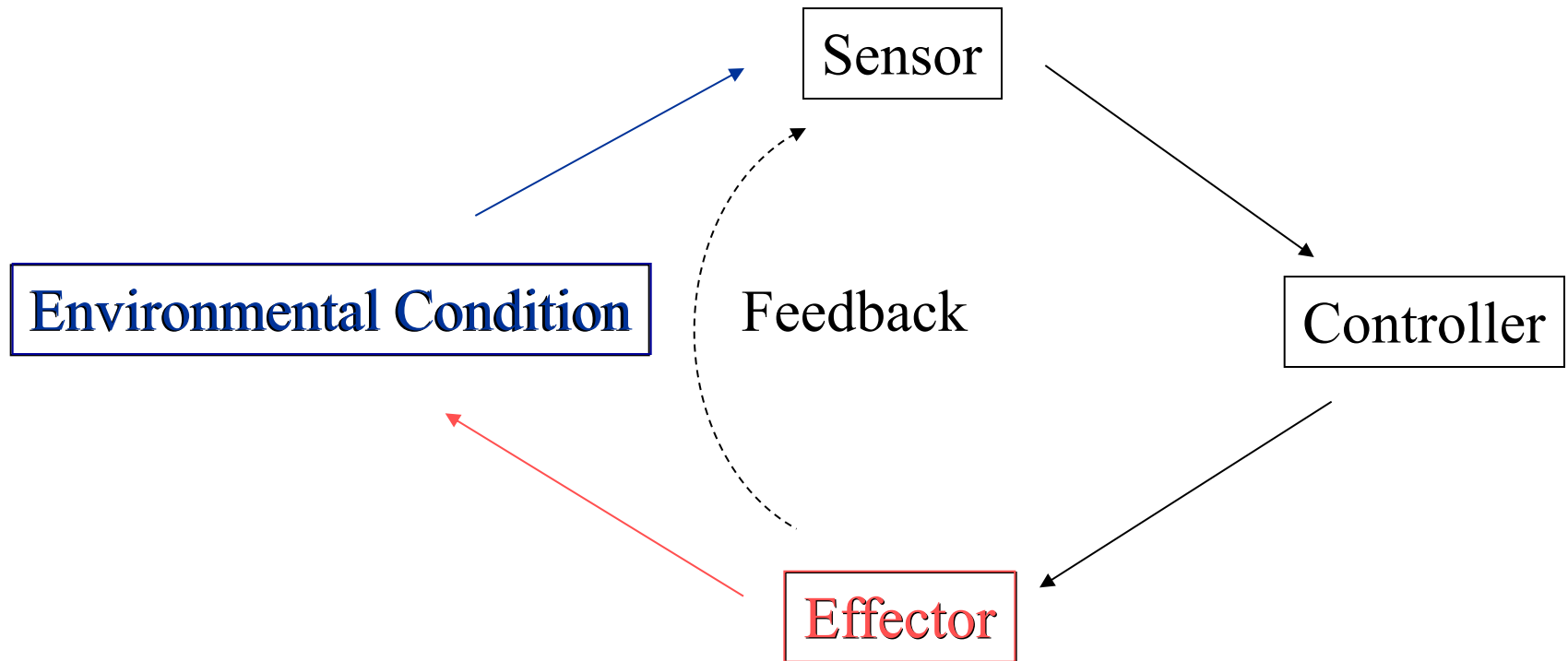
Regulatory Mechanism Components

- *sensor*
 - monitors internal conditions, detects changes
- *integrating center (controller)*
 - receives & integrates information
- *effector*
 - responds to changes
 - activity of effectors results in return of condition to normal levels.

Regulatory Mechanisms Work by Feedback

- *Feedback* = return of output to the input part of a system
- The response of the effector influences subsequent output by the effector

Feedback



Types of Homeostatic Responses

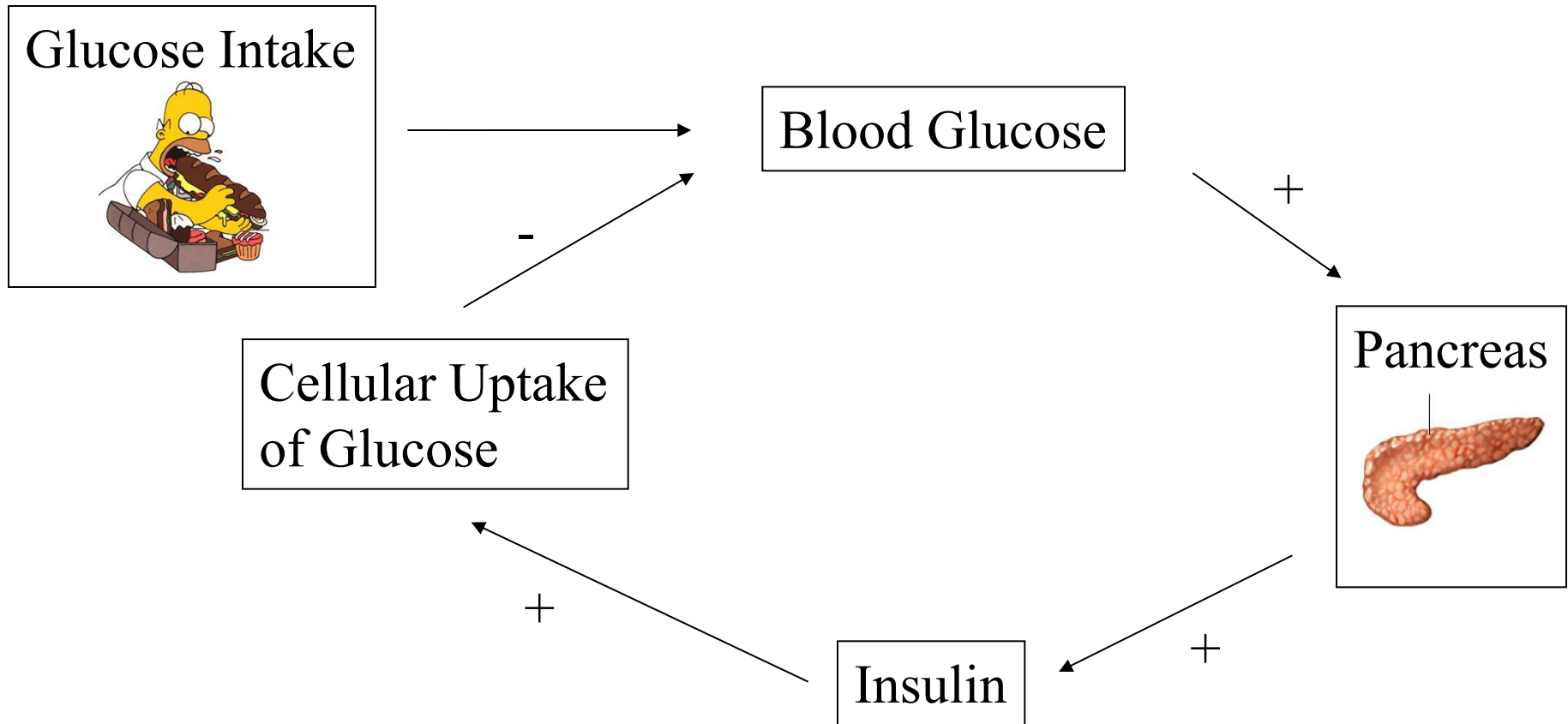
- **Negative Feedback**

- change in a condition leads to a response from the effector *which counteracts* that change

↑ Change → ↑ Response → ↓ Change → ↓ Response

- most common type of response

Negative Feedback Example: Insulin in Blood Glucose Regulation



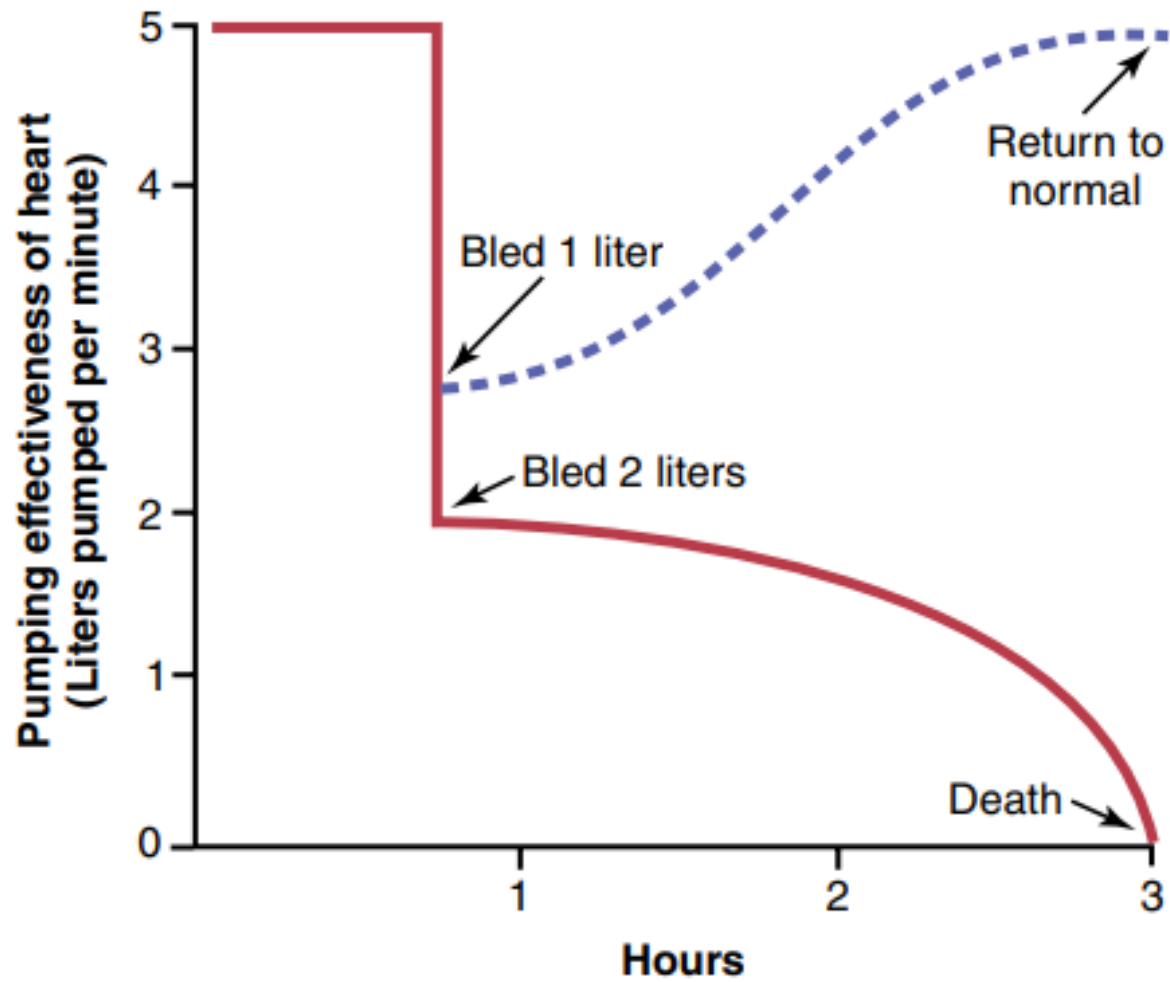
Types of Homeostatic Responses

- **Positive Feedback**

- change in a condition leads to a response from the effector which *amplifies* that change

↑ Change → ↑ Response → ↑↑ Change → ↑↑ Response

- less common, and part of a larger negative feedback loop



Applied Physiology is the study of biological systems and steps into practice.

It involves the application of the knowledge of **physiological** properties to restore core **stability** and joint **stability**. It **differs** from **clinical practice**.

Examples

- Space physiology
- Deep sea physiology
- High altitude physiology
- Advanced exercise physiology

Means slightly different physiology mechanisms but still normal

Clinical practice

- Treatment of diseased people
- And **abnormal**

Back to normal stage = balance

Teaching Methods:

- Lectures,
- Problem solving,
- Small group discussions,
- Case based education.

Take home message

So.....

Please know **what is normal first at Earth level....then**

Apply it to different environments like
space level or submarine level

Then.....

Come to **clinical aspects** disorders and
treatment

- MBS 210 – physiology ++++ RS, CVS,
GIT, REP
- MBS 310 - physiology
- MBM 320 – physiology ++++
- MBM 220 - physiology
- MBP/MBE 260- Anat & Physiology +++

Human physiology credits

Contact hours:

- ☐ Lectures - 6 hours per week,
- ☐ Practicals - 3 hours every 2 weeks
- ☐ Tutorial - 1 hour per week
- ☐ Seminars - 3 hours every 2 weeks

Mean = 10 hrs a week

Total = 400 contact hrs/subject

Medical Attributes

- Self directed
- Tenacity
- Autodidactic
- Professionalism
- Dignity
- No Bias/ Non judging = objectivity
- Eclectic

In mbchb...



degree

Prescribed books

Ganong's Review of Medical Physiology (Review Questions). by Kim E Barrett; Susan M Barman; Jason X -J Yuan; Heddwen Brooks;. 26 edition **eBook**. English. **2022**

2. Walter F.B., Boupaep E.L., (2009) Medical Physiology. Publishers Saunders/Elsevier. ISBN 978-1-4160-3115-4

Recommended book

Purchase **Guyton** and
Hall **Textbook of Medical
Physiology** 2022- 14th Edition. Print **Book&
E-Book**. ISBN 9781455770052,
9780323389587.

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Da Vinci Code of mind mapping

Neurophysiology of mindset

Neurophysiology of club drugs

Author name : Balapala

